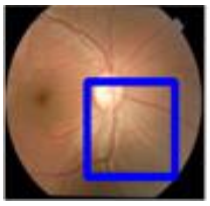
Digital Image Processing Lab (NCSH 302) **Session: Monsoon 2026-27**
B. Tech. 5th Sem (CSE-Hons) **Date: 04.08.2026**

Practice Assignment (Basics of Image Processing)

1. Write a Python program to read the "**ratina.png"** image and resize it to 200 × 200 pixels, then perform the following tasks:

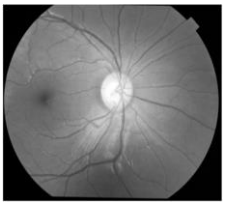
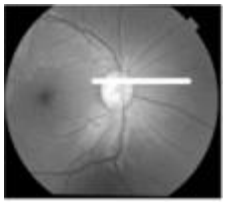
    a. Draw a blue rectangular region of interest (ROI) between the points (80, 80), (80,160), (160, 80), and (160, 160) with a thickness of 5 pixels. Then display the ROI marked image using matplotlib.

    Sample Input Image: **Sample Output Image:**

    b. Convert the image into a grayscale image, then draw a white straight line from the point (80, 80) to the point (160, 80), then display the image.

    Sample Input Image: **Sample Output Image:**

2. Write a program to take a grayscale image $\boldsymbol{I}(\boldsymbol{X},\boldsymbol{Y})$ as the input, then perform the following tasks without using the library function:

    a. Obtain the image $\boldsymbol{I}'(\boldsymbol{X},\boldsymbol{Y})$ by rotating the image $\boldsymbol{I}(\boldsymbol{X},\boldsymbol{Y})$ 90 degrees in clockwise direction.

    Sample Input Image: **Sample Output Image:**

b. Display the vertically mirror image of $\boldsymbol{I(X,Y)}$, and save that mirror image in the working directory.

Sample Input Image:

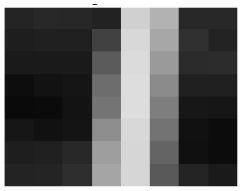


Sample Output Image:



3. A grayscale image is stored as a 2D matrix, where each element represents the intensity of a pixel (0-255). Write a program to extract an **8 × 8** patch from the grayscale image of **"Natural_scene.jpg"** image. Apply **binary thresholding** with a threshold value of **50**, such that pixels with intensity **less than 50** are assigned **0**, and pixels with intensity **greater than or equal to 50** are assigned **1**. Display both the original and thresholded patches. Implement the thresholding operation manually using **nested loops** without using any built-in thresholding functions.

Sample Input Image:



Sample Output Image:



4. A digital image is stored as a 2D matrix, each element of the matrix is known as a pixel. A binary image is a type of digital image in which every pixel is represented either by $0\ or\ 255$. Write a program which will take a binary image as input and will check whether any $\mathbf{2 \times 2}$ subpart of that image is a copy of the top left $\mathbf{2 \times 2}$ subpart of that same image. If yes then print $\boldsymbol{True, X}$ and if not then print $\boldsymbol{Faslse.\ X}$ is the number of times the top left $\mathbf{2 \times 2}$ subpart is copied within the whole image.

Sample Input Image:

0	255	255	0
255	0	0	255
0	0	255	0
255	255	255	0

Sample Output: $True, 1$